



Evaluating area-level features for proxy means test models: evidence from rural, semi-urban and urban districts in poverty targeting

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Abstract

While satellite imagery and administrative records have been widely used for poverty mapping, their role in poverty targeting, particularly within the Proxy Means Test (PMT) framework, remains underexplored. This study evaluates the usefulness of area-level features, such as Night-Time Light intensity, vegetation and built-up indices (NDVI, NDWI, NDBI), land cover types and the number of schools and health facilities, in improving household-level expenditure estimates and reducing targeting errors in PMT models. Using household data from a variety of district types in Indonesia, including rural, semi-urban and urban districts, we assess how these features affect model performance in terms of inclusion and exclusion errors under both 20% and 40% poverty thresholds. Our findings show that area-level features improve PMT model performance primarily in districts with a high number of sub-districts, particularly in semi-urban settings. In contrast, in urban and rural districts with limited sub-district granularity, these features often introduce noise or degrade accuracy. We also observe systematic misclassification patterns: poor households in more developed sub-districts tend to be wrongly excluded, while non-poor households in lower resource areas are wrongly included. Using the Shapley value approach, we show that the predictive effect of features such as NDVI varies by context, for instance, in some districts, NDVI positively correlates with expenditure due to productive croplands, diverging from broader trends. These results underscore the importance of spatial granularity, threshold sensitivity and local context when integrating area-level data into poverty targeting systems.

Keywords Proxy means test · Poverty targeting · Area-level features · Satellite imagery · Administrative records · Inclusion error · Exclusion error

1 Introduction

Poverty targeting remains a crucial aspect of social policy in many countries, aiming to allocate limited resources effectively to those most in need. Among the widely adopted methods for poverty targeting is the Proxy Means Test (PMT), which estimates monthly per capita household expenditure (*pcexp*) in the following period. Households are ranked based on estimated *pcexp* from the highest to the lowest. An eligibility threshold, such as 40% in Indonesia, is applied so that all households in the bottom 40% are classified as poor or eligible for social assistance. A PMT model is evaluated not only by estimation error but also by targeting errors comprising inclusion error (IE) and exclusion error (EE). IE is the proportion of non-poor households amongst those predicted poor. EE represents the proportion of the poor households which are predicted as non-poor. When using a known fixed threshold T that determines the proportion of households classified as poor (or eligible), the total number of households predicted as poor remains constant. In this case, every incorrectly excluded household (*false negative*) must be replaced by an incorrectly included household (*false positive*), ensuring that inclusion and exclusion error is the same.¹ The PMT is favoured for its relative simplicity and reliance on easily observable household indicators rather than direct income measures, which are often difficult to obtain. However, the method has notable limitations. Estimating *pcexp* through the PMT relying only on survey data has long been subject to high inclusion and exclusion errors [11].

The rise of new data sources (non-survey data), such as earth observation data, has introduced opportunities to enhance model accuracy. Satellite imagery provides a rich source of data that captures real-time economic and environmental conditions across vast regions. For instance, Night-Time Light (NTL) intensity, which serves as a proxy for economic activity, has been widely utilised in poverty estimation models. NTL intensity is frequently found as the most important variable in predicting the change of region wealth [16, 18, 21, 26, 28], as an alternative to asset-based poverty indices derived from survey data at region level [17]. However, Xu et al. [29] found that some districts with similar NTL intensity values have obviously different poverty indices, and NTL intensity suffers from the issue of over-glow [26] where light can spill over from one cell to an adjacent cell. Previous studies explored the potential of non-survey data that is primarily from satellite imagery for poverty *mapping*. This work studies the usefulness of features collected from satellite imagery and administrative records at **sub-district level** (hereafter called **area-level features**) for poverty *targeting*. We investigate whether area-level features can enhance the accuracy and robustness of PMT models in estimating monthly per

¹ More precisely, using standard terminology for true positive (TP), false positive (FP), and false negative (FN), and where $| \cdot |$ denotes set cardinality, IE is defined as $|FP|/|TP \cup FP|$ and EE is defined as $|FN|/|TP \cup FN|$, where the positive (poor) class is $|TP \cup FN|$. In the case where the predicted poor and poor sets are the same size, i.e. $T\%$ of the households, so $|TP \cup FP| = |TP \cup FN|$. This means that $|FP| = |FN|$, hence $IE=EE$.

capita household expenditure at **district level** across varied geographic and socio-economic conditions.

Our previous study [27] highlighted that while additional sub-district features consisting of vegetation indices and the number of education and health clinics have the potential to improve model predictions, they can sometimes lead to worse performance in certain districts due to regional disparities and data noise. We showed that the inclusion of Normalised Difference Vegetation Index (NDVI), Normalised Difference Built-up Index (NDBI), Normalised Difference Water Index (NDWI) and the number of schools and health clinics collected at sub-district level improves PMT performance in Surabaya (urban district) and Brebes (semi-urban district), yet it does not improve PMT models in Sumba Tengah (rural district). A possible explanation for this is that for the added features to be useful, they need to enable further discrimination among the data relevant to expenditure estimation. Area-level features, vegetation indices and the number of schools and health clinics in Sumba Tengah tend to be very similar in all sub-districts, adding noise to models. We found that area-level features work differently in different districts. This underscores the need for a deeper understanding of how, when and why these features influence model estimation and accuracy.

In this paper, we expand our previous work with more area-level features derived from earth observation data via satellite imagery and more selective area-level features collected from administrative records. We study whether area-level features can reduce *pcexp* estimation error and poverty targeting errors across rural, semi-urban and urban districts in Indonesia at poverty targeting thresholds 20% and 40%. This broader scope enables us to examine the effects of area-level features across diverse regional contexts, with distinct socio-economic characteristics. We investigate whether sub-district-level features, comprising NTL intensity, NDVI, NDWI, NDBI, area of each most common land cover type, number of public and private schools, and number of public health clinics, can discriminate the poverty status of households **within** districts as opposed to **between** districts, as early research discussed the differences between districts. Previous research [10, 16–18, 20, 21, 24–26] used area-level features as proxies for measurement to distinguish between areas. In our setting, we look at the usefulness of sub-district features **within** districts.

The results show that NTL intensity, NDVI, NDWI, NDBI, land cover types and number of schools and health clinics provide spatial and socio-economic characteristics of sub-districts. Values of these features vary across sub-districts and across years, increasing variances in household survey data and allowing PMT models to distinguish households within districts, specifically between sub-districts. However, the usefulness of sub-district features in minimising poverty targeting errors as measured by IE/EE is sensitive to the number of sub-districts and the threshold used to define poor and non-poor households. We find that the inclusion of area-level features improves model performance, measured by reductions in both IE and EE, only in districts with a relatively high number of sub-districts. In our datasets, this pattern is consistently observed in semi-urban districts, which typically have more than 15 sub-districts. In contrast, rural and urban districts, which generally have fewer than 10 sub-districts, show no consistent improvement and in many cases, experience increased error rates when area-level

features are added. This suggests that spatial granularity is a critical factor: when the number of sub-districts is too low, area-level features may introduce noise or fail to capture meaningful variation in household-level poverty status.

We further investigate the types of households which are prone to be misclassified by PMT models with area-level features that are potentially discriminatory. We find that the high IE/EE produced by PMT models with area-level features results from systematic errors rather than random errors. The monthly per capita household expenditure of poor households is systematically overestimated, especially for poor households in wealthy areas. Wealthy areas are frequently signalled by high NTL intensity, high NDBI and the presence of private schools, while poor areas are frequently indicated by low NTL intensity, low NDBI and the absence of private schools. Use of household features such as per capita floor space and asset ownership is likely to underestimate the *pcexp* of non-poor households. Non-poor households who live in small spaces in or near to a business centre are prone to be incorrectly classified as poor. To prevent the impact of this on the application of poverty targeting, we recommend performing bias correction post-processing. A post-modelling correction layer could be introduced that adjusts predictions based on the likelihood of overestimation. For example, households with low reported monthly per capita consumption expenditure living in high NTL intensity or urban areas could be flagged for additional scrutiny or be prioritised for further verification through field visits or follow-up surveys.

We employ the Shapley value approach to examine how monthly per capita household expenditure is estimated by PMT models and to assess the contribution of area-level features. The Shapley value approach, derived from game theory, assigns a value to each “player” (or feature) by considering all possible combinations of players [12, 22]. Extending this approach to the comparative usefulness of features, the Shapley value calculates a value for each feature based on its contribution across all possible feature combinations in the model. This enables comparisons across models and helps assess the competitive influence of different subsets of features. To reveal how these estimations are made, we use the SHAP (Shapley Additive Explanations) package in Python [13].

This paper is organised as follows: Sect. 1 introduces poverty targeting and the Proxy Means Test; Sect. 2 presents a review of the literature on poverty mapping and poverty targeting that uses features derived from satellite imagery; Sect. 3 shows that area-level features improve *pcexp* estimations and reduce IE/EE when using XGBoost models to implement the Proxy Means Test across rural, semi-urban and urban districts in Indonesia; Sect. 4 includes a detailed analysis of the impact of area-level features on the PMT models, including a study of feature importance using Shapley values; Sect. 5 provides a conclusion and discussion of further work.

2 Background

The Proxy Means Test (PMT) has become the favoured poverty targeting method compared to other methods such as the Demographic and Health Survey (DHS) wealth index, the Poverty Probability Index (PPI), Community Based Targeting

(CBT) and Livelihood Empowerment Against Poverty (LEAP). The PMT performs better than standard methods when defining the poor based on a purchasing power parity (PPP) of US\$2 per capita consumption and better than the CBT method in predicting the poor based on poverty lines [1]. This is aligned with findings by Poulin et al. [19] who developed PMT models using machine learning algorithms and found that machine learning based PMT outperforms other poverty targeting methods at screening out households having assets associated with wealth. Machine learning based PMT is also deemed to be fairer by community members than other methods [19].

The PMT has relied on survey data for many years. A survey, which requires large-scale fieldwork, has often been the primary source for constructing poverty indices. However, such data is inherently limited by temporal delays, coverage gaps and high cost. To overcome these constraints, recent studies have explored the integration of non-survey data—primarily satellite imagery features—alongside machine learning techniques to create more granular and timely poverty maps. In this work, we use features from satellite imagery and administrative records, and calculate feature values at sub-district level. We investigate the use of area-level features from the literature that have been found to be relevant to poverty mapping.

Night-Time Light (NTL) intensity along with multi-angle NTL intensity have high coefficient of determination R^2 in the estimation of poverty rate at cluster level [24]. NTL intensity captures infrastructure development, urbanisation and household wealth distribution, making it an invaluable non-survey variable. For some circumstances, using NTL intensity alone as the poverty predictor is insufficient. Xu et al. [29] found that some districts with similar NTL intensity values have obviously different poverty indices. As noted above, NTL intensity suffers from the issue of over-glow [26] where light can spill over from one cell to an adjacent cell. For example, it is difficult to find a dark neighbourhood in a city due to the spillover of light from bright neighbourhoods. However, NTL intensity combined with other variables such as traffic accessibility, altitude, impervious surface coverage, flat area coverage, road density, cropland coverage and risk index show the capability of capturing the spatial variance in poverty [29]; NTL intensity combined with daylight time images in a transfer learning scheme also can estimate consumption expenditure and asset wealth at cluster level [10, 28].

Other indices derived from daytime satellite imagery such as Normalised Difference Vegetation Index (NDVI), Normalised Difference Water Index (NDWI) and Normalised Difference Built-Up Index (NDBI) have been shown useful for poverty mapping at finer granular levels, e.g. $10 \times \text{km}^2$ grids. NDVI indicates the density of plant growth and higher values indicate a higher density of green vegetation [23]; NDWI [8] is sensitive to changes in liquid water content of vegetation canopies, and is complementary to, but not a substitute for NDVI, while NDBI is designed to differentiate built-up urban areas from woodland or farming areas [30]. Dawson et al. [7] showed that poverty related areas are significantly correlated with positive high and/or negative NDVI in both shrinking and growing cities. NDVI has positive correlation with poverty rate at sub-district level [21] although Putri et al. [20] showed that NDVI and NDWI have a weak correlation with official poverty rates at district (regency) level. The combination of NTL intensity with vegetation indices NDVI,

NDBI and NBWI can alleviate the phenomenon of NTL brightness in vegetated regions and increase the variability of light intensity values in the urban core [31] where these features increase the accuracy of built-up areas monitoring. Utami et al. [25] developed Spatially Granular Poverty Index (SGPI) in urban areas by combining NDVI, NDBI, NDWI and other satellite imagery features and showed that SGPI has high correlation with poverty rate at 1 km resolution.

Another non-survey data source commonly used for poverty mapping is land cover data. Land cover data enables detection of changes in land use such as changes in arable land, built-up land, water bodies and unused land. Land cover data potentially reflects the living condition of households. Barbier and Hocard [5] showed that households on remote less-favoured agricultural lands are susceptible to poverty-environment traps. Ge et al. [9] showed that arable land continuously decreased while built-up land and vegetation gradually expanded. They also showed that there are obvious regional differences of land use change overtime. Significant differences exist between deep poverty-stricken counties and normal poverty-stricken counties on arable land and built-up change, indicating more intense human construction activities in normal poverty-stricken areas [9]. Shao and Li [24] showed that multi-angle NTL intensity and ratio of each land cover type can accurately estimate poverty rate at province level.

To improve the accuracy of poverty estimation models, it is essential to identify key features that can explain the characteristics of an area and distinguish one area from another. These features often reflect physical and environmental factors such as infrastructure, land use, proximity to resources, and the presence of natural elements like forests or bodies of water. By incorporating such area-level features in poverty estimation models, we aim to capture broader patterns that household-level features alone might miss. However, introducing too many area-level features can introduce noise into the model, especially when the household survey data is small. Therefore, in this work, we identify and evaluate informative area-level features while keeping their number minimal to prevent overfitting and maintain model robustness.

We provide a method to collect features from satellite imagery and administrative records at area level that can capture the physical and environmental factors of an area, indirectly relating to economic development and poverty. The area-level features consist of total annual NTL intensity, total area based on land cover type, Normalised Difference Vegetation Index (NDVI), Normalised Difference Water Index (NDWI), Normalised Difference Built-Up Index (NDBI), access to education (number of public schools) and access to health services (number of public health clinics). NTL intensity unambiguously distinguishes between the most and the least poor quintiles, but is weak to separate middle quintiles. NTL intensity show substantial and moderate agreement with assets-based indices [17]. The average intensity of NTL is the most contributing feature in a model predicting poverty head count at sub-district level [21] and the most important feature of a model predicting Multidimensional Poverty Index (MPI) at commune level [18]. There is a link between household wealth and the level of night emissions [26]. Other features such as road access, access to markets, labour force participation, etc., are potentially used in poverty mapping. However, those features are either not available at sub-district level for all time periods examined in this study or are not reliable.

3 Proxy means tests with area-level features

3.1 Datasets

We categorise datasets into household-level and area-level datasets. Household-level datasets are from Indonesia's National Socio-economic Survey, known as SUSENAS, which is performed twice a year with different sample sizes depending on the required estimation level. SUSENAS variables used in PMT modelling comprise demographic variables such as the number of toddlers, adults and elderly in the household, marital status and educational attainment; employment variables such as number of household members working in the agriculture, industry and service sectors and labour status for each household member in each workforce category; house condition consisting of types of roof, wall and floor materials, type of sanitation and source of drinking water; asset ownership variables comprising ownership status of house, television, fridge, water heater, boat, motorbike, car and jewellery; and target variable monthly total consumption expenditure. Monthly per capita household expenditure is derived from the monthly total of food and non-food consumption expenditure divided by the number of household members. There are a total of 60 features with low variance among households in each district. We use five years SUSENAS data from 2016 to 2020 collected from 21 districts in three large provinces in Indonesia—Jawa Tengah, Jawa Timur and Nusa Tenggara Timur—covering rural, semi-urban and urban districts. Table 1 presents the number of households sampled, number of sub-districts and district type. As shown in Table 1, the number of households sampled is proportional to the population size, the more the number of sub-districts, the more households are sampled.

Table 2 provides a description of area-level features used in this study. All area-level features are calculated at sub-district level which equates to administrative unit level 3.² The total area of each land cover type, total of annual NTL intensity, mean of median NDVI values, mean of median NDBI values and mean of median NDWI values are collected from satellite imagery and calculated through Google Earth Engine. For land cover classification, we apply the Land Cover Type 1—Annual International Geosphere-Biosphere Programme (IGBP) classification, which defines 17 land cover types. Given our focus is Indonesia, a tropical country, we use only the seven most common land cover types for rural and semi-urban districts comprising evergreen broadleaf forests, woody savannas, savannas, croplands, urban/built-up areas, cropland/natural vegetation mosaics and water bodies, and use only the four most common land cover types for urban districts consisting of croplands, urban built-up, natural vegetation and water. The total area for each land cover type at the sub-district level is calculated for year $t - 1$ and linked to the survey data from year t .

The median values of NDVI, NDWI and NDBI for each month from 2015 to 2020 are calculated from Landsat 8 Surface Reflectance (L8 SR) images through

² Note that in Indonesia, province (*provinsi*) is LAU 1, district/city/regency (*kota/kabupaten*) is LAU 2 and sub-district (*kecamatan*) is LAU 3.

Table 1 Dataset sizes

District	Name	2016–2019	2020	Sub-districts	District Type
1	Kota Blitar	1937	543	3	Urban
2	Kota Madiun	2027	547	3	Urban
3	Kota Salatiga	2057	549	4	Urban
4	Kota Pasuruan	2048	565	4	Urban
5	Kota Surakarta	2643	691	5	Urban
6	Sumba Tengah	1685	532	5	Rural
7	Sumba Barat	1804	547	6	Rural
8	Kota Kupang	2345	649	6	Urban
9	Nagekeo	1926	532	7	Rural
10	Lembata	1785	543	9	Rural
11	Rote Ndao	1788	525	9	Rural
12	Sumba Barat Daya	2061	544	11	Rural
13	Belu	2038	563	12	Rural
14	Madiun	2998	822	15	Semi-urban
15	Brebes	3764	1010	17	Semi-urban
16	Purbalingga	3020	814	18	Semi-urban
17	Ngawi	3151	850	19	Semi-urban
18	Tuban	3318	902	20	Semi-urban
19	Pati	3303	904	21	Semi-urban
20	Sumenep	3301	906	26	Semi-urban
21	Kota Surabaya	4286	1082	31	Urban

Google Earth Engine. To avoid anomalous values and bias due to cloud contaminated pixels, the L8 SR images are filtered using a cloud masking technique that aims to remove cloud and cloud shadow contaminated pixels through Level 1 Band pixel quality assessment. Then the median value of each index is calculated from the preprocessed L8 SR images for each month based on each sub-district's polygon. These median values are then averaged over the 12-month period from March to February in the following year (this period being chosen because SUSENAS surveys are done in March). The values are aggregated to provide environmental indices for each year at sub-district level.

The last eight area-level features are from administrative records which are published annually by Statistics Indonesia (*Kabupaten Dalam Angka*). Different from other research, we prefer to not use OpenStreetMaps to collect information about the number of healthcare clinics and schools because there are inconsistencies and missing data for the majority of sub-districts and time periods examined in this study. Access to services is a key indicator of wealth at the sub-district level as the availability of services makes an area more desirable, driving up prices (such as housing) and thereby increasing the average wealth of residents who can afford to live there. We include features representing two types of healthcare services and six out of the 17 available education services at the sub-district level, expressed as numerical values reflecting the number of each service in a given area. For healthcare services, we

Table 2 Description of area-level features

Data	Source	Area level	Resolution
Total area of each land cover type	MODIS Land Cover Type (MCD12Q1) Version 6.1	Sub-district	500 ms
Total of annual NTL intensity	Annual Global VIIRS	Sub-district	463.83 ms
Mean of median NDVI	Landsat 8 Surface Reflectance (Band 4 Red & Band 5 Near Infrared)	Sub-district	30 ms
Mean of median NDBI	Landsat 8 Surface Reflectance (Band 5 Near Infrared & Band 6 SWIR)	Sub-district	30 ms
Mean of median NDWI	Landsat 8 Surface Reflectance (Band 5 Near Infrared & Band 6 SWIR)	Sub-district	30 ms
Number of public healthcare	Administrative records	Sub-district	NA
Number of public maternity and child healthcare	Administrative records	Sub-district	NA
Number of public elementary school (Public ES)	Administrative records	Sub-district	NA
Number of public junior high school (Public JHS)	Administrative records	Sub-district	NA
Number of public senior high school (Public SHS)	Administrative records	Sub-district	NA
Number of private elementary school (Private ES)	Administrative records	Sub-district	NA
Number of private junior high school (Private JHS)	Administrative records	Sub-district	NA
Number of private senior high school (Private SHS)	Administrative records	Sub-district	NA

consider public healthcare (*puskesmas*) and maternity and child healthcare (*posyandu*). In terms of education services, we focus on three key services to ensure data consistency and avoid sparsity. We find that the remaining education services only present in a limited number of sub-districts.

3.2 Experimental setup

We develop PMT models using a machine learning algorithm, eXtreme Gradient Boosting (XGBoost) [6]. We choose XGBoost because it outperforms other methods such as Ridge regression and Random Forest regression [16, 27, 29]. We perform hyperparameter tuning for each district independently and for both feature sets used, household features only and household plus area-level features. All hyperparameter tuning uses data from 2016 to 2019, and applies Grid Search with 5-fold cross validation with negated mean squared error as the evaluation metric. Hyperparameters tuned comprise the learning rate, maximum tree depth, number of trees (estimators) and regularisation parameter (λ).

The best hyperparameter values for each combination of district and variable set are used to develop PMT models. For each district, we develop two PMT models, a PMT model with household features only and a PMT model with household and area-level features. Each PMT model uses data from 2016 to 2019 as training set and 2020 data as test set. We evaluate all PMT models across 21 districts and report all results on the test set. We also evaluate all scenarios under two poverty thresholds: 20% and 40%. That is, households ranked in the bottom 20% or 40% are classified as poor, while the rest are considered non-poor. We do not use poverty lines to define the poor as most poverty targeting methods adopt ranking based methods, such as Indonesia's Subsidised Rice Program (known as *Raskin*) targeting the poorest 30% [3], Indonesia's Conditional Cash Transfer program (known as *PKH*) targeting the poorest 5–10% [2] and Indonesia's Non-Cash Food Assistance targeting the poorest 25% [4].

3.3 Results

We present the results comparing PMT models without and with area-level features using XGBoost regressor. Table 3 presents the RMSE of XGBoost models using household-level features (HH) and those augmented with area-level features (HH+Area), evaluated across a variety of district types. This analysis uses the same model framework employed to assess inclusion error (IE) and exclusion error (EE). The results reinforce a key finding: the usefulness of area-level features is strongly tied to spatial granularity. In semi-urban districts with a higher number of sub-districts, such as Ngawi, Brebes and Sumenep, the addition of area-level features reduces RMSE, indicating improved accuracy in household expenditure estimation. However, performance gains are not guaranteed even with sufficient granularity.

In districts Tuban and Pati, with 20 and 21 sub-districts respectively, the models show negligible or even negative change, suggesting that spatial granularity is necessary but not sufficient. Other factors such as intra-district heterogeneity or

Table 3 RMSE ($\times 1000$) Rupiah of XGBoost models with household (HH) features only and with household and area-level (HH+Area) features, train: 2016–2019, test: 2020

District	Type	Sub-districts	HH	HH+Area
Kota Blitar	Urban	3	1486	1469
Kota Madiun	Urban	3	1138	1131
Kota Salatiga	Urban	4	1679	1622
Kota Pasuruan	Urban	4	715	704
Kota Surakarta	Urban	5	1331	1342
Sumba Tengah	Rural	5	385	375
Sumba Barat	Rural	6	490	470
Kota Kupang	Urban	6	861	884
Nagekeo	Rural	7	412	392
Lembata	Rural	9	639	625
Rote Ndao	Rural	9	554	529
Sumba Barat Daya	Rural	11	332	333
Belu	Rural	12	476	460
Madiun	Semi-urban	15	567	565
Brebes	Semi-urban	17	555	538
Purbalingga	Semi-urban	18	824	820
Ngawi	Semi-urban	19	504	485
Tuban	Semi-urban	20	1137	1150
Pati	Semi-urban	21	777	779
Sumenep	Semi-urban	26	373	359
Kota Surabaya	Urban	31	1832	1909

Bold indicates better performing model

the informativeness of the features may be influencing results. Modest improvements are also observed in rural districts with at least seven sub-districts, such as Belu and Lembata. In contrast, urban districts with only three to five sub-districts consistently show little benefit from area-level features, likely due to limited spatial variation. These findings provide important context for interpreting patterns of targeting errors IE/EE discussed in the following section.

Figure 1 shows the performance of XGBoost models measured by IE/EE at poverty threshold 20%. In rural districts, XGBoost with household and area-level features (XGBoost+HH+Area) often performs similarly or slightly better than XGBoost with household features (XGBoost+HH) in most districts. However, in a few districts (e.g. Sumba Barat Daya), the addition of area-level features increases the error. This shows that area-level features sometimes help, but inconsistently across rural districts. XGBoost+HH+Area outperforms XGBoost+HH across most semi-urban districts. However, in most urban districts, inclusion of area-level features makes models perform worse than models without area-level features, except in Kota Surabaya. Kota Surabaya differs from other urban districts because it has more sub-districts (31 sub-districts) compared to others (less than 10 sub-districts). In this case, area-level features are beneficial for Kota Surabaya but ineffective for others as the diversity in data provided by area-level features is limited.

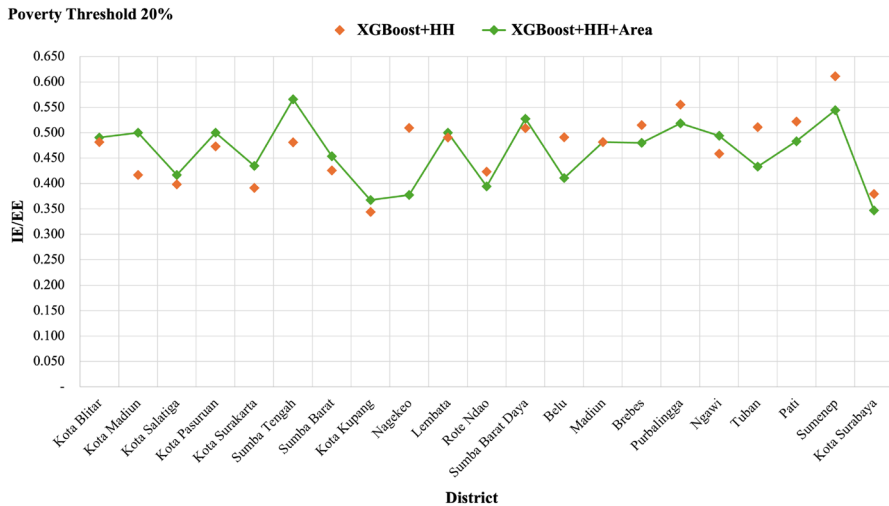


Fig. 1 Inclusion and exclusion errors (IE/EE) at poverty threshold 20% of XGBoost with households features only (XGBoost+HH) and XGBoost with household and area-level features (XGBoost+HH+Area), train: 2016–2019, test: 2020

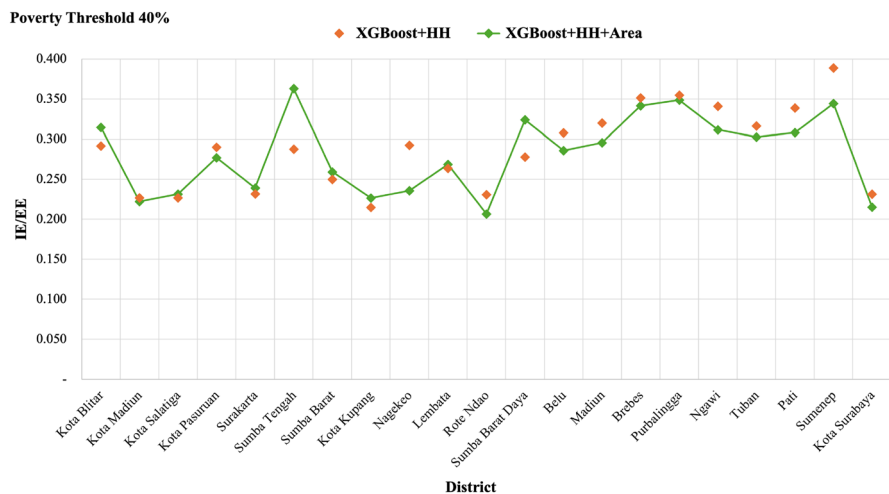


Fig. 2 Inclusion and exclusion errors (IE/EE) at poverty threshold 40% of XGBoost with households features only (XGBoost+HH) and XGBoost with household and area-level features (XGBoost+HH+Area), train: 2016–2019, test: 2020

Figure 2 shows IE/EE of XGBoost models without and with area-level features at poverty threshold 40%. At this threshold, XGBoost+HH+Area performs better or the same as XGBoost+HH in almost all districts. IE/EE is significantly lower than at the threshold 20%. This indicates that the model benefits more from area-level features when the poverty threshold is more inclusive. In semi-urban districts,

the performance of XGBoost+HH and XGBoost+HH+Area is comparable, with small differences across districts. There is a slight trend where XGBoost+HH+Area slightly improves or remains stable, implying that area-level features are more useful when targeting a broader group. In urban districts, at this higher threshold, XGBoost+HH+Area performs comparably or better, particularly in districts 17–21 (Ngawi to Kota Surabaya). Similar to semi-urban districts, area-level features become more beneficial when the model targets a broader population in urban contexts.

Adding area-level features improves PMT model performance only in districts with many sub-districts. In this study, these improvements are consistently observed in semi-urban districts, which have higher sub-district counts compared to rural or urban districts. In contrast, rural and urban districts, which typically have fewer sub-districts, show no improvement or even a slight degradation in performance when area-level features are included, likely due to insufficient spatial variation or added noise.

This finding suggests that spatial granularity is a critical factor: when the number of sub-districts is too low, area-level features may introduce noise or fail to capture meaningful heterogeneity, thereby reducing their effectiveness. Districts Pati (21 sub-districts) and Tuban (20 sub-districts), both semi-urban, show consistent reductions in IE/EE at both 20% and 40% poverty thresholds when area-level features are used. In contrast, districts Rote Ndao (9 sub-districts, rural) and Kota Blitar (3 sub-districts, urban) show no improvement or worsened errors, illustrating the limited utility of area-level data in low-granularity contexts. At the 20% poverty threshold, improvements in IE/EE are more pronounced, especially in semi-urban districts. This indicates that area-level features are effective in identifying the poorest households when there is enough spatial resolution. At the 40% poverty threshold, IE/EE improvements are more visible in the same districts. At this threshold, area-level features also help prevent under-coverage when targeting broader segments of the population. Overall, district structure (e.g., number of sub-districts), spatial granularity and targeting intensity (poverty threshold) influence the utility of area-level features in poverty targeting models.

4 Analysis of area-level features

4.1 Correlation between area-level features and poverty indicators

Before assessing the usefulness of area-level features sourced from satellite imagery and administrative records in the Proxy Means Test (PMT) models, we first investigate the correlations between these features and poverty indicators at the sub-district level. Specifically, we explore whether these correlations: (1) are consistent across districts, (2) exhibit strong positive or strong negative relationships, and (3) can indicate the feature's potential to improve PMT models. The poverty indicators analysed include the median of monthly per capita expenditure and the share of the population in the poorest 20% at district level.

Figure 3 displays correlations between area-level features and the median of monthly per capita household expenditure at the sub-district level across 21 districts. In rural districts, particularly those with fewer sub-districts, Night-Time Light (NTL) intensity shows consistently strong positive correlations with median monthly per capita household expenditure, making it a reliable proxy for economic activity in less developed districts. Vegetation and land cover features (e.g. NDVI, NDWI, evergreen broadleaf, savannas) tend to have strong negative correlations across semi-urban districts, indicating that the greener or more forested regions often correspond to lower expenditure levels. However, this trend is not universal as some districts, such as Sumenep, show the opposite pattern, where these features correlate positively with median monthly per capita household expenditure. This suggests that the interpretation of vegetation features is context-dependent. Education and healthcare infrastructure show mixed or weak correlations in these rural districts. As the number of sub-districts increases in rural districts, the correlations become more varied, reflecting greater internal diversity and complexity in these areas.

In semi-urban and urban districts, especially those with many sub-districts, education infrastructure, particularly private schools, emerges as a strong positive indicator of higher expenditure levels. Meanwhile, the predictive power of NTL intensity weakens or even reverses, likely due to saturation effects in highly electrified urban settings. Vegetation-related features remain negatively correlated across both semi-urban and urban districts, although their strength varies. Urban built-up area shows weaker or inconsistent patterns, particularly in densely developed districts. Overall, districts with more sub-districts tend to produce more stable and interpretable correlations, suggesting that a greater number of sub-districts improves the reliability of area-level feature associations with economic indicators.

Figure 4 displays correlations between area-level features and the proportion of the poorest 20% at the sub-district level. In rural districts, particularly those with

District	Kota Bitar	Kota Madiun	Kota Salatiga	Kota Pasuruan	Kota Sukakarta	Sumba Tengah	Sumba Barat	Kota Kupang	Nagekeo	Lembata	Rote Ndao	Sumba Barat Daya	Belu	Madiun	Brebes	Purbalingga	Ngawi	Tuban	Pati	Sumenep	Kota Surabaya
NTL	1.00	-0.50	0.40	0.20	0.60	0.90	0.94	-0.09	0.07	0.33	0.10	0.41	0.55	0.35	0.75	0.50	0.04	0.57	0.58	-0.01	0.13
Public healthcare																					
Maternity and child healthcare	1.00	-0.50	-0.60	-0.20	0.60	-0.56	0.60	-0.60	0.02	-0.08	-0.35	-0.07	-0.11	-0.23	0.55	0.10	0.17	0.05	-0.11	-0.36	-0.14
Public ES	0.87	-0.87	0.21	0.00	0.80	-0.82	0.26	-0.76	-0.13	0.15	-0.41	-0.24	-0.15	-0.28	0.32	0.03	-0.30	0.37	0.23	-0.15	0.11
Public JHS	1.00	-1.00	1.00	0.63	0.95	-0.97	0.41	-0.84	0.09	0.16	0.01	-0.08	-0.36	-0.06	0.16	0.24	-0.46	0.23	0.18	0.07	0.05
Public SHS	0.00	-0.50	0.63	-0.32	0.21	0.00	0.21	-0.06	0.18	0.18	0.27	0.00	0.46	0.25	0.25	0.39	-0.48	-0.01	0.31	0.02	0.19
Private ES	0.00	0.50	0.11	-0.40	0.10	-0.60	0.12	0.26	-0.02	0.41	0.27	-0.06	-0.19	-0.01	0.21	0.05	-0.02	0.35	0.17	0.37	0.06
Private JHS	0.87	-1.00	0.63	-0.40	0.87	0.22	0.62	0.47	0.25	0.11	0.55	0.23	0.73	-0.17	-0.19	0.28	-0.04	0.37	0.24	-0.03	0.01
Private SHS	1.00	0.50	0.95	-0.80	0.80	0.71	0.83	0.00	-0.08	0.37	0.55	0.44	0.69	-0.34	-0.13	0.16	-0.10	0.26	0.13	0.39	-0.05
NDVI	0.50	0.50	-0.80	1.00	0.50	0.60	-0.37	0.03	-0.18	0.43	0.27	0.35	-0.51	0.35	-0.60	-0.41	-0.56	-0.45	-0.49	0.17	0.24
NDBI	-0.50	-0.50	0.60	-0.20	-0.60	-0.60	0.49	0.83	0.32	-0.22	-0.02	-0.27	0.38	0.45	0.60	0.34	0.49	-0.08	0.55	-0.32	-0.07
NDWI	0.50	0.50	-0.60	0.20	0.60	0.60	-0.49	-0.83	-0.32	0.22	0.02	0.27	-0.38	-0.45	-0.60	-0.34	-0.49	0.08	-0.55	0.32	0.07
Evergreen broadleaf			0.95			-0.10	-0.60	-0.43	0.22	-0.03		0.45	-0.46	-0.42	-0.47	-0.27	-0.60	-0.14	-0.55	0.23	
Woody savannas			0.20			-0.50	-0.83	-0.78	-0.29	0.60	0.47	0.20	-0.57	-0.38	-0.50	-0.46	-0.49		-0.50	0.26	
Savannas	-0.50	0.50	0.40	0.20	0.22	-0.70	-0.66	-0.66	-0.07	-0.03	0.32	-0.19	-0.17	-0.25	-0.36	-0.18	-0.65	0.29	-0.30	0.34	-0.06
Croplands	0.50	0.50		-0.11		-0.70	0.49	-0.83	0.36	-0.10	-0.42	0.05	0.02	0.02	0.58	0.47	0.51	-0.07	0.15	-0.25	-0.22
Urban built-up	-0.50	-0.50	0.00	0.80	0.60	-0.50	0.93	0.09				0.29	0.72	0.17	0.66	0.49	0.12	0.37	0.27	0.16	0.04
Natural vegetation	1.00	0.50	0.20	1.00	0.22	-0.50	0.14	-0.82	-0.13	-0.68	-0.35	-0.36	-0.20	0.40	0.05	0.07	-0.30	-0.53	-0.60	0.10	-0.27
Water				-0.80		-0.41	-0.52	-0.39	-0.18	-0.20	-0.43	-0.63	0.13	0.22	0.46			0.42	0.23	0.31	-0.20
Precipitation	0.50	-0.50	0.40	0.80	0.50	0.80	0.89	-0.83	-0.36	0.17	0.15	0.26	-0.64	-0.47	-0.57	-0.31	-0.50	-0.44	-0.25	-0.01	-0.15

Fig. 3 Correlation between area-level features and median monthly per capita household expenditure in year 2020

fewer sub-districts, many area-level features show strong and consistent negative correlations with the proportion of the poorest 20%, especially education and infrastructure variables. For example, NTL intensity, private school availability and urban built-up area generally exhibit strong negative correlations. This suggests that better infrastructure and services are associated with a lower proportion of the poorest households. Conversely, some natural features such as NDVI, NDWI and water coverage occasionally show positive correlations, indicating that greener or water-abundant areas may still host larger shares of the poorest, possibly due to limited economic diversification or accessibility.

In semi-urban and urban districts, especially those with more sub-districts, patterns become more nuanced. The negative correlation between education infrastructure and poverty remains but is often weaker and more variable. Private junior and senior high schools still correlate negatively with poverty share in some districts, but less consistently than in rural zones. Interestingly, natural features such as vegetation and croplands begin to show more positive correlations, implying that in urban contexts, greener areas may paradoxically coincide with higher poverty concentrations—perhaps reflecting informal settlements near green belts or peri-urban zones. Districts with a higher number of sub-districts also show more stable and interpretable patterns, reinforcing the idea that increased granularity improves the robustness of correlations between features and poverty indicators. Correlation patterns shown in Figs. 3 and 4 suggest that both the type and scale of geographic units influence the effectiveness of area-level features for poverty targeting, and that different sets of features may be more appropriate depending on the context.

4.2 Spatial distribution of area-level features

Before we further analyse how area-level features are used to estimate monthly per capita household expenditure (*pcexp*) by XGBoost, we discuss the

District	Kota Blitar	Kota Madiun	Kota Salatiga	Kota Pasuruan	Kota Sukakarta	Sumba Tengah	Sumba Barat	Kota Kupang	Nageko	Lembata	Rote Ndao	Sumba Barat Daya	Belu	Madiun	Brebes	Purba-lingga	Ngawi	Tuban	Pati	Sumenep	Kota Surabaya
NTL	-0.50	-0.29	0.20	0.00	-0.60	0.40	-0.83	-0.14	-0.64	-0.70	-0.52	-0.58	-0.69	-0.50	-0.34	-0.82	-0.64	-0.45	-0.82	-0.27	-0.15
Public healthcare		-0.04	0.45	0.63	-0.67	-0.32	-0.28	0.15			0.00	0.52	0.10		-0.36	-0.18	-0.52	-0.37	-0.34	-0.15	0.45
Maternity and child healthcare	-0.50	-0.21	0.80	0.80	-0.60	-0.87	-0.94	0.26	-0.25	-0.72	0.10	0.54		-0.50	-0.08	-0.09	-0.58	-0.39	-0.03	-0.29	0.55
Public ES	0.00	0.25	0.11	0.80	-0.80	0.10	-0.55	0.76	-0.22	-0.50	0.14	0.31	0.38	0.00	-0.24	0.11	-0.49	-0.15	-0.48	-0.28	0.20
Public JHS	-0.50	0.22	-0.80	0.32	-0.95	-0.21	-0.41	0.55	-0.31	-0.32	-0.05	0.13	0.65	0.50	-0.08	0.31	-0.22	-0.41	-0.29	-0.11	0.06
Public SHS	0.87	-0.41	-0.32	0.95	-0.21	0.87	-0.21	-0.32	-0.39	-0.55	-0.55	-0.39	-0.21	-0.50	-0.28	-0.25	-0.37	0.11	-0.53	-0.40	-0.19
Private ES	-0.87	0.14	-0.63	0.40	-0.10	0.50	-0.68	0.06	-0.14	0.03	-0.73	0.11	0.32	-1.00	-0.05	-0.23	-0.43	-0.47	-0.32	0.05	0.32
Private JHS	-0.87	-0.08	-0.95	0.40	-0.87	-0.45	-0.93	-0.79	-0.44	-0.84	-0.55	-0.21	-0.80	0.50	0.19	-0.20	-0.46	-0.15	-0.36	0.27	0.26
Private SHS	-0.50	-0.08	-0.74	0.20	-0.80	-0.35	-0.83	0.00	-0.43	-0.73	-0.55	-0.31	-0.69	0.50	0.53	-0.25	-0.58	-0.26	-0.40	0.22	0.17
NDVI	0.50	0.46	1.00	-0.40	-0.50	0.60	0.26	0.26	0.32	0.47	0.40	-0.03	0.83	0.50	0.47	0.73	0.07	0.32	0.34	0.49	-0.67
NDBI	-0.50	0.03	-0.80	0.80	0.60	-0.60	-0.60	-0.60	-0.29	-0.45	0.40	-0.03	-0.61	-0.50	-0.54	-0.63	0.18	-0.32	-0.48	-0.41	0.63
NDWI	0.50	-0.03	0.80	-0.80	-0.60	0.60	0.60	0.60	0.29	0.45	-0.40	0.03	0.61	0.50	0.54	0.63	-0.18	0.32	0.48	0.41	-0.63
Evergreen broadleaf		0.61	-0.95			0.60	0.49		0.21	0.48	0.28	-0.07	0.54		0.47	0.47	0.15	0.26	0.47	0.27	
Woody savannas		0.37	0.40			0.40	0.49	0.54	-0.43	0.20	-0.27	0.08	0.69		0.48	0.75	0.04		0.58	0.36	
Savannas	1.00	0.52	0.20	-0.80	-0.22	0.00	0.09	0.66	-0.36	-0.40	-0.57	0.16	0.45	0.50	0.47	0.13	0.16	-0.57	0.32	0.13	-0.32
Croplands	0.50	-0.04		-0.63		-0.70	-0.14	0.54	-0.80	-0.84	0.18	-0.27	0.06	0.50	-0.20	-0.46	-0.36	-0.05	-0.43	-0.14	-0.26
Urban built-up	-0.50	-0.07	0.40	0.20	-0.60	-0.70	-0.64	-0.37				-0.34	-0.63	-0.50	-0.51	-0.66	0.06	-0.59	-0.01	-0.14	-0.25
Natural vegetation	-0.50	-0.35	-0.40	-0.40	-0.22	-0.60	0.43	0.88	-0.58	-0.43	-0.12	0.48	0.52	0.50	0.59	-0.48	-0.04	0.31	0.39	-0.27	-0.11
Water		0.49		-0.20		-0.05	0.75	0.15	-0.29	-0.53	0.72	0.49	-0.22		-0.52		-0.49	0.00	-0.02	-0.01	
Precipitation	0.50	0.49	0.00	0.20	-0.50	0.30	-0.54	0.89	0.68	0.43	-0.33	0.06	0.71	-0.50	0.57	0.59	-0.21	0.67	-0.02	0.36	-0.19

Fig. 4 Correlation between area-level features and proportion of the poorest 20% in year 2020

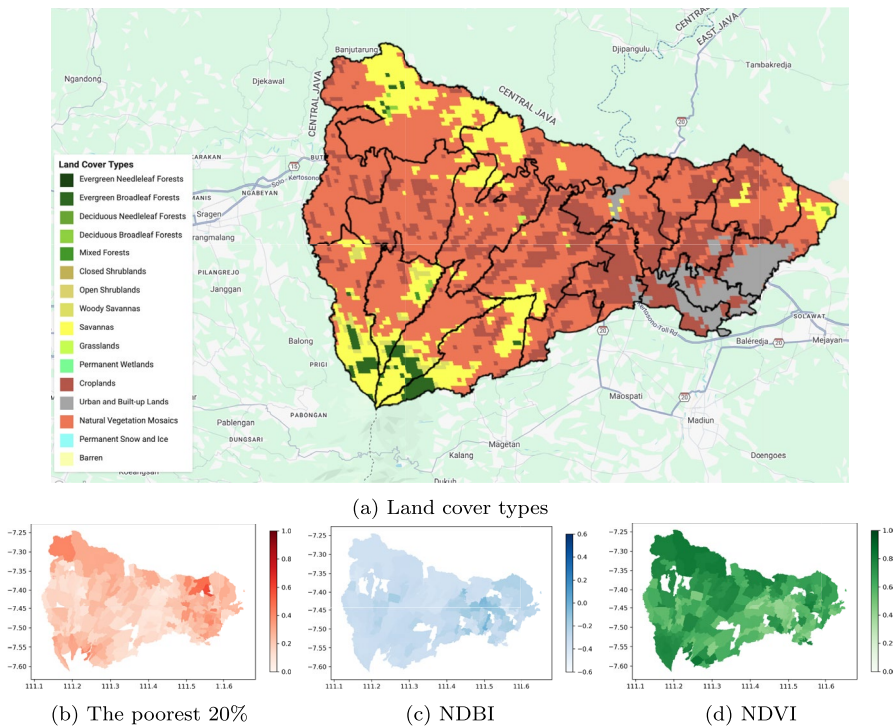


Fig. 5 Distribution of land cover types, the proportion of the poorest 20%, NDBI and NDVI in 2020 in district Ngawi

distribution of some area-level features in district Ngawi. Ngawi is chosen for analysis because the correlations in Fig. 3 differ for some features from those of comparable districts. With both poverty thresholds (40% and 20%), Ngawi shows a moderate discrepancy between the performance of XGBoost+HH and XGBoost+HH+Area. At 40% threshold, IE/EE of XGBoost+HH+Area (IE/EE = 0.312) is lower compared to XGBoost+HH (IE/EE = 0.341), however, at 20% threshold, although both models are comparable, XGBoost+HH slightly outperforms XGBoost+HH+Area, raising questions about when additional area-level features fail to improve poverty targeting. This makes Ngawi a compelling case to explore why area-level features might not add value, despite the assumption that they should. Ngawi is neither too small nor too large in terms of number of sub-districts. It allows for sufficient granularity in analysis without being skewed by either too few or too many regions, making it ideal for testing generalizability.

The spatial distribution of land cover types, poverty, NDBI and NDVI shown in Fig. 5 reveals consistent patterns that help explain the socio-environmental landscape of the district. Urban and built-up areas, identified in grey on the land cover map (Fig. 5a), are primarily located in the southeastern part of Ngawi. These areas also exhibit high values of NDBI (Fig. 5c), which is indicative of

dense built environments, and relatively low NDVI values (Fig. 7c), reflecting limited vegetation cover. Interestingly, these same areas correspond to regions with a lower proportion of the poorest 20% (Fig. 5b), suggesting that urbanisation is associated with improved economic conditions. In contrast, the central and western regions, dominated by croplands (brown) and grasslands (yellow), have low NDBI, higher NDVI, and a noticeably higher concentration of poor households. This pattern highlights the vulnerability of rural and agricultural zones despite the vegetative richness.

The relationship between vegetation, built environments and poverty levels becomes clearer when comparing the NDVI and NDBI patterns with the poverty distribution. While areas with high NDVI suggest healthy vegetation—mostly associated with croplands and grasslands—these do not necessarily correlate with lower poverty. In fact, some of the most vegetated areas also contain the highest proportions of the poorest 20%, particularly in the south-central and southwestern parts of the district. This means that natural vegetation or agricultural land alone may not guarantee economic well-being, likely due to limited access to infrastructure, services or market opportunities. On the other hand, regions with higher NDBI values, despite having low vegetation, tend to show lower poverty rates, implying that built infrastructure may play a more direct role in supporting livelihoods and reducing poverty. These patterns underscore the importance of considering both environmental and infrastructural features when analysing poverty dynamics at the area level.

4.3 Impact of area-level features

The improvement (reduction in targeting errors IE/EE) is sensitive to the number of sub-districts as the level of spatial disaggregation. The level of spatial disaggregation is used to collect the features from both satellite imagery and administrative records. In this section, we further analyse how the estimation of monthly per capita household expenditure (*pcexp*) is made by XGBoost when it includes area-level features. We inspect the contribution of area-level features to classification errors, specifically non-poor households classified as poor (inclusion error IE) and poor households classified as non-poor (exclusion error EE) using a 20% poverty threshold in district Ngawi.

The majority of exclusion errors are concentrated in sub-districts I and J, where 10 poor households in each are wrongly classified as non-poor. Interestingly, these sub-districts are among the most developed, with higher annual NTL intensity, more private schools and more public maternity and child healthcare centres compared to others, as detailed in Table 4. This suggests that despite overall development, these areas contain heterogeneous household conditions, and area-level features may fail to capture pockets of poverty, leading to under-targeting.

Meanwhile, inclusion errors—non-poor households misclassified as poor—are more prevalent in sub-districts B and D. Sub-district B shows 14 inclusion errors, double the number of its true poor households (7), and sub-district D shows 9 inclusion errors, despite having 19 poor households. These areas are less affluent than I and J but still show over-targeting, possibly due to misleading signals from

Table 4 Sub-district characteristics in district Ngawi, number of poor and non-poor households and number of misclassified poor and non-poor households

Sub-District	NTL	Public School			Private School			Healthcare		True		Misclassified as	
		ES ^a	JHS ^b	SHS ^c	ES ^a	JHS ^b	SHS ^c	Public	MC	Poor	Non-poor	Poor	Non-poor
A	227	30	3	1	3	2	0	1	78	6	24	3	4
B	187	31	3	1	2	2	0	1	73	7	43	14	3
C	175	26	2	1	1	2	1	1	63	9	21	1	6
D	218	27	3	1	1	0	0	1	65	19	30	9	3
E	424	27	2	1	1	0	1	1	74	5	45	2	3
F	181	21	1	0	1	1	0	1	42	7	33	1	3
G	197	21	2	1	0	1	1	1	45	3	27	2	1
H	115	22	2	0	0	0	0	1	42	5	35	7	1
I	1202	31	6	1	5	8	3	2	107	19	51	7	10
J	467	44	3	1	2	1	1	2	93	17	62	3	10
K	458	39	4	1	3	6	1	2	91	25	55	5	8
L	113	20	4	0	0	0	0	1	50	5	25	5	2

Bold indicates sub-districts discussed in the text

^a Elementary School

^b Junior High School

^c Senior High School

^d Public Maternity and Child Healthcare

area-level proxies or insufficiently nuanced model behaviour. Together, these findings highlight that area-level features can distort classification in both directions, missing the poor in well-developed areas and falsely identifying poverty in moderately developed ones. These findings also underscore the importance of refining how these features are integrated into PMT models.

Figure 6 shows the impact of each feature on *pcexp* estimation, ordered by feature importance, defined using Shapley value. Although area-level features have the same values for households within the same sub-district, several of these features still significantly influence the model's estimations. Notably, six area-level features comprising number of public elementary schools, sum of NTL intensity, NDVI, cropland area, NDBI and number of private elementary schools, stand out as particularly impactful among the 97 features evaluated. An interesting pattern emerges with the “number of public elementary schools” feature, where fewer public elementary schools in a sub-district is associated with higher estimated *pcexp*. This trend aligns with Table 4, which shows that most households incorrectly classified as poor are from sub-districts with a high number of public elementary schools. Additionally, NTL intensity follows a pattern consistent with existing research, where higher NTL intensity correlates with higher estimated *pcexp*, indicating urban development. Furthermore, Fig. 6 suggests that district Ngawi has an agricultural focus. This inference is drawn from observed relationships: larger cropland areas correspond with higher *pcexp* estimates, lower NDVI values (often indicating less vegetation density or non-crop areas) are linked to

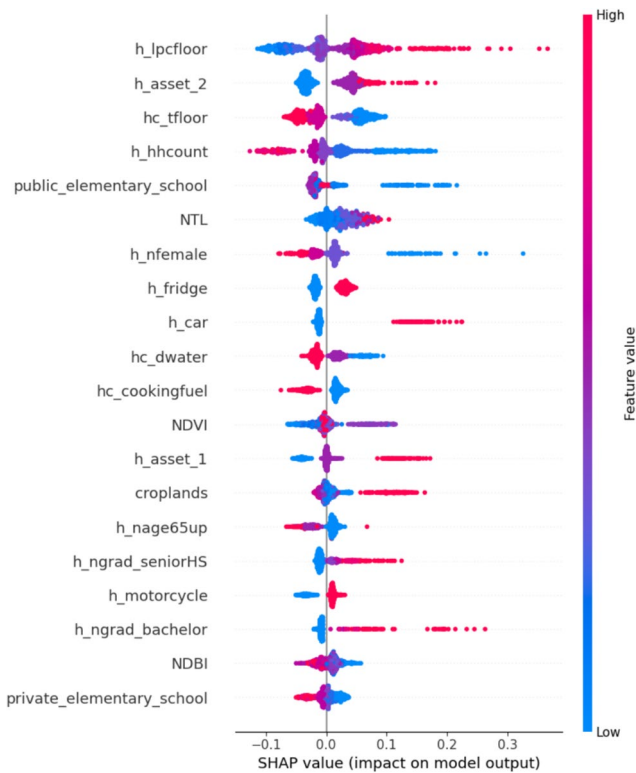


Fig. 6 Impact of feature values on estimated *pcexp* by XGBoost model at threshold 20% in district Ngawi

lower *pcexp* estimates, and lower NDBI values (indicating fewer built-up areas) contribute to higher *pcexp* estimates.

Household-level features also play an essential role, with per capita floor space (*h_lpcfloor*) emerging as one of the most influential features in the model's output. Larger floor space generally leads to higher *pcexp* estimates, suggesting that this feature strongly correlates with wealth. Features such as household head count (*h_hhcount*) and asset ownership (e.g. refrigerator, car) similarly impact estimates, where lower values tend to reduce *pcexp* estimates, often contributing to the classification of households as poor. Additionally, access to basic amenities, such as improved cooking fuel and water sources, increases *pcexp* estimates, aligning with indicators of higher living standards.

Distinctive patterns in the area-level features also reveal important insights into misclassification tendencies. For instance, the presence of private schools is associated with higher *pcexp* estimates, while sub-districts with more public elementary schools often see lower estimates. This pattern may lead the model to misclassify households in sub-districts with a higher number of public schools as poorer than they actually are, resulting in overestimation for households in sub-districts with private schools or more developed amenities. Additionally, the

reliance on NTL intensity as an indicator of development supports higher *pcexp* estimates in urban areas, but may overstate wealth levels for some households in rural sub-districts with high NTL intensity. These insights suggest that while the model effectively incorporates both household and area-level features, refining the weights or interactions between urbanisation and educational features could improve accuracy, particularly in distinguishing between genuinely poor and non-poor households in diverse socio-economic settings.

We closely examine the contribution of area-level features to the estimations for households that were wrongly included and excluded from poverty targeting. Fig. 7a–d illustrate the estimated *pcexp* in natural log (y-axis) and the impact of each feature in either increasing or decreasing *pcexp* estimates across all wrongly excluded poor households, ordered by sub-district codes. In these figures, red bars indicate features that increase *pcexp* estimates, while blue bars show features that decrease *pcexp* estimates.

The use of natural log for *pcexp* as the output variable standardises these effects. The width of each bar represents the contribution (Shapley value) of each feature to the estimation by the XGBoost model. The sum of NTL intensity, total cropland area, mean NDVI and the number of public elementary schools are among the top 15 most important features in the XGBoost model. The base value shown in these figures is 603K (natural log 13.31), reflecting the average monthly *pcexp* in the training set (2016–2019 data). In contrast, the average monthly *pcexp* and the average estimated monthly *pcexp* by XGBoost in the test set (2020 data) for this district are 876K(13.68) and 725K(13.49), respectively. These results indicate that the XGBoost model tends to underestimate *pcexp* in 2020, as *pcexp* in the training set is lower than in the test set. Notably, sum of NTL intensity consistently increases *pcexp* estimates for nearly all wrongly excluded households, aligning with Table 4, which shows that most excluded households are from sub-districts with higher sum of NTL intensity than other sub-districts.

Since NTL intensity serves as an indicator of development or urbanisation, these findings suggest that poor households (at the 20% threshold) residing in urbanised areas have their *pcexp* overestimated, leading to their misclassification as non-poor. The total area of croplands, mean NDVI and the number of public elementary schools also contribute to increased *pcexp* estimates in specific sub-districts. In sub-districts I and J, both the total area of croplands and mean NDVI follow a similar pattern, indicating that households in these areas may rely on agriculture. In these sub-districts, higher cropland areas and NDVI values correlate with increased *pcexp*, suggesting a positive relationship between agricultural activity and expenditure in these locations. This observation contrasts with general findings across other districts and prior literature, where NDVI is commonly negatively associated with household expenditure. The results in Ngawi highlight that NDVI can reflect different economic conditions depending on land use, such as productive agricultural croplands versus forested or underdeveloped areas. Estimations at the household level reveal an interesting insight: area-level features can reflect the dominant economic sector or job type of households within the area, providing context for *pcexp* estimates based on local characteristics.

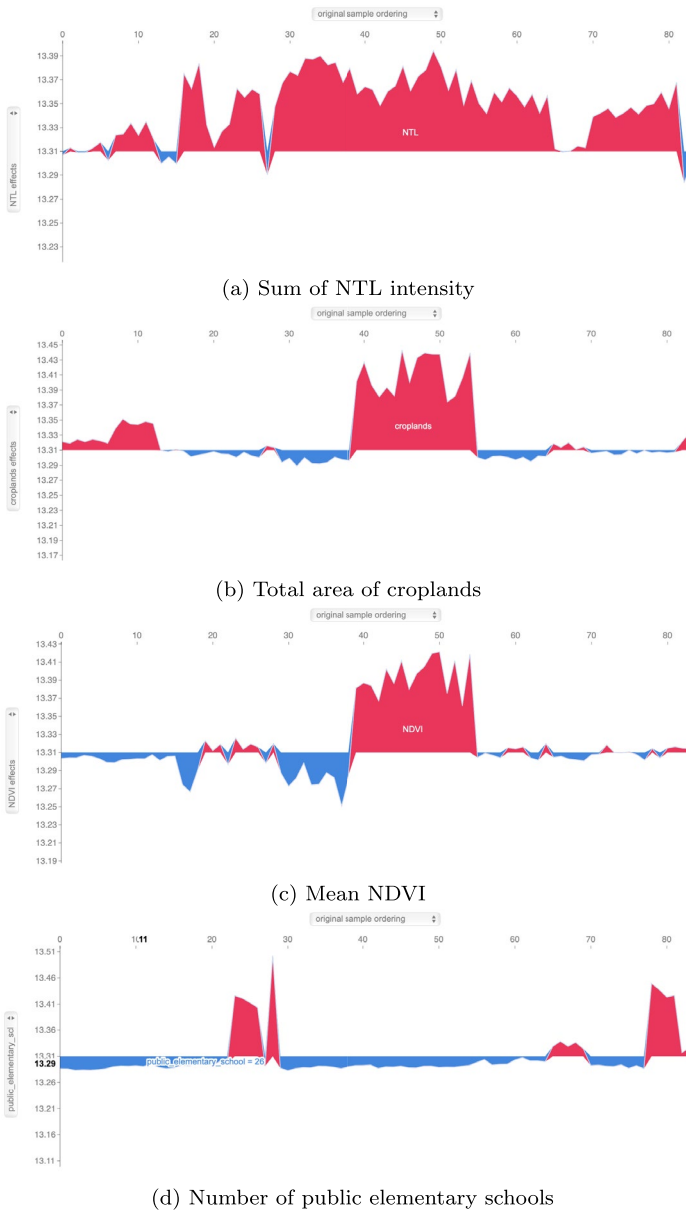


Fig. 7 Impact of area-level features on *pcexp* estimations of wrongly excluded households (exclusion error) in district Ngawi by XGBoost model

4.4 Analysis of misclassification errors

We further examine the contribution of these features to (*pcexp*) estimation for some of the households misclassified by the XGBoost model. We closely look

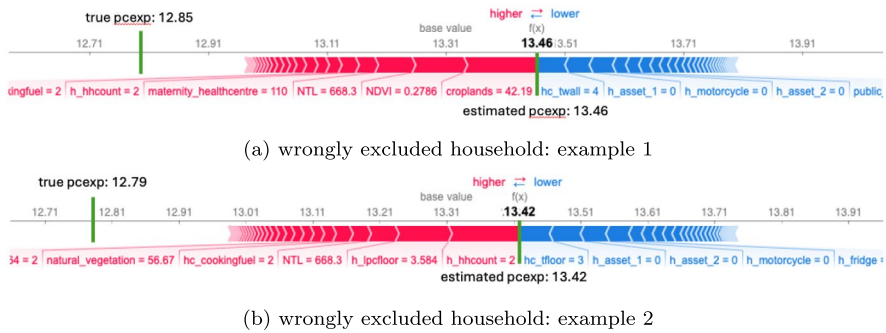


Fig. 8 Poor households classified as non-poor households in district Ngawi where the inclusion of area-level features does not reduce IE/EE at threshold of 20%

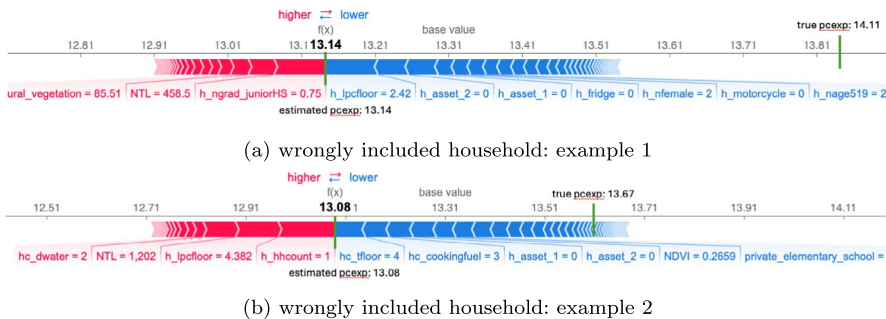


Fig. 9 Non-poor households classified as poor households in district Ngawi where the inclusion of area-level features do not reduce IE/EE at threshold of 20%

at individual cases of wrongly excluded (EE) and wrongly included (IE) households. Figure 8 illustrates the feature contributions for two examples of poor households that were misclassified as non-poor. In Fig. 8a, area-level features such as the total area of croplands, mean NDVI, sum of NTL intensity and the number of child and maternity health clinics centres contribute significantly to increasing the *pcexp* estimates, while features related to wall type and household assets reduce the *pcexp* estimates. In Fig. 8b, the sum of NTL intensity and the area of natural vegetation are two of the primary factors driving the increase in *pcexp* estimates. Both households have estimated *pcexp* values higher than the average estimated *pcexp* in the district. These figures demonstrate that households situated in sub-districts with high NTL intensity values, exceeding those in other sub-districts, are at risk of having their *pcexp* overestimated. This overestimation affects the household ranking, which may lead to the exclusion of genuinely poor households from the bottom 20% poverty threshold.

Figure 9a shows the feature contributions for wrongly included households (non-poor households classified as poor). In Fig. 9a, household-level features such as per capita floor space and asset ownership play a significant role in

reducing the *pcexp* estimates, leading to underestimation. Additionally, area-level features such as higher NDVI values and the absence of private schools further contribute to lowering the *pcexp* estimates. Figures 8 and 9 show that in district Ngawi, area-level features such as NTL intensity, as well as the presence of private schools and healthcare centres, generally increase *pcexp* estimates. This indicates that these area-level features can influence the model's estimation patterns in this district.

In both figures, household-level features such as per capita floor space, number of assets, and type of water and cooking fuel have a notable impact on the *pcexp* estimation, with limited floor space and a lack of assets (e.g. refrigerator or motorcycle) pulling the estimates down. While area-level features such as NTL intensity, which signals urbanisation, and higher NDVI values, generally associated with rural areas, contribute slightly to increasing the estimates, they are not enough to counteract the strong negative impact of the household-level features. The absence of private schools, often an indicator of lower income areas, further reinforces the model's tendency to underestimate *pcexp* in these cases. This pattern suggests that household-specific features may have a stronger influence than area-level features, leading to an overall underestimation and resulting in the misclassification of non-poor households as poor.

4.5 Further insights and discussion

Understanding how features contribute to *pcexp* estimations leads to several valuable insights. First, we find that underestimation and overestimation of *pcexp* result from systematic errors rather than random ones. Households with low monthly per capita expenditure in wealthier areas tend to have their *pcexp* overestimated. Indicators such as NTL intensity and access to private schools and healthcare signal an area's wealth or urbanisation level. Households in these wealthier or urbanised areas may have access to better job opportunities, but they also face higher costs for housing, food and services. As a result, they generally have higher expenditures than households in less urbanised or less affluent areas. This helps explain why poor households in wealthier areas are often misclassified as non-poor, as the model associates higher area-level expenditures with individual households.

Another systematic error we observe is that underestimations are primarily influenced by household-level features, such as per capita floor space and the absence of assets. Per capita floor space can introduce noise; for example, households living in or near business centres often have less floor space due to high accommodation costs or limited housing availability. Many of these households choose to rent smaller spaces, such as rooms instead of houses, even if their expenditures are relatively high. Similarly, the absence of assets, particularly transportation assets such as cars or motorcycles, strongly influences the model's predictions. Households near business centres may not own vehicles because they can rely on public transport to commute to work, school, healthcare facilities and grocery stores. However, a lack of transportation assets can cause the model to classify these households as poorer than they actually are. Additionally, the model's treatment of asset ownership

introduces limitations. Asset-related questions typically capture only ownership status, not the quantity or quality of assets. For instance, households with one motorcycle are considered the same as those with two or three motorcycles, even though owning multiple vehicles is common in Indonesia. This simplification may lead to inaccurate poverty classifications, as the model cannot distinguish between households with differing levels of asset wealth.

Secondly, land cover type can serve as an indicator of the local job sector, which in turn correlates with household expenditure. As noted in the previous discussion, in certain sub-districts, larger areas of cropland and higher NDVI values are associated with increased *pcexp* estimates by the model. This suggests that agriculture-related land cover types and vegetation indices such as NDVI do not always correspond to poorer areas. Households involved in the agricultural sector are not necessarily low-expenditure households; in fact, they may be wealthier than others. Areas with extensive and fertile cropland (reflected in high NDVI values) can contribute positively to household wealth, as these lands create job opportunities and economic benefits for households residing in the vicinity.

This study equips policymakers and National Statistics Offices (NSOs) such as Statistics Indonesia, with an evidence-based procedure for designing better poverty surveys. By ranking both households and area-level variables according to their real contribution to PMT accuracy, this study allows NSOs to choose only questions that are useful in the model for estimation. This translates into shorter questionnaires, less interviewer time, smaller budgets and faster data turnaround. At the same time, our feature-attribution analysis shows how misclassification occurs, for example, the *pcexp* of poor households in bright, service-rich neighbourhoods is often overestimated, while the *pcexp* of non-poor households in cramped urban flats is typically underestimated. Therefore, policymakers and NSOs can employ sample bias correction checks, for instance, flagging households with high NTL but low reported expenditure, into their statistical process.

Equally important, our study shows how non-survey data can plug the gaps that budget-constrained surveys leave behind. NTL intensity, vegetation and built-up indices, land cover mix and counts of schools or clinics, harvested from satellite imagery and routine administrative records, reduce inclusion and exclusion errors in districts that have enough sub-districts to capture spatial variation. Because these features can be frequently updated, NSOs can update poverty eligibility lists in near-real time and postpone expensive resampling until model error drifts above a policy threshold. Together, these insights let NSOs build a hybrid data pipeline that is cheaper to run, lighter on field staff and more accurate.

5 Conclusion and further work

This study demonstrates that area-level features derived from satellite imagery and administrative records, such as Night-Time Light (NTL) intensity, vegetation indices (NDVI, NDWI), built-up index (NDBI), land cover types and the number of schools and health clinics, can enhance Proxy Means Test (PMT) models by improving estimates of monthly per capita household expenditure and reducing poverty targeting

errors. Many of these features, particularly NTL intensity and school-related indicators, rank among the top contributors to model predictions. By introducing additional variance into the data, area-level features help distinguish between households within and across sub-districts, which is especially valuable in districts with a higher number of sub-districts.

However, we found that the utility of area-level features is sensitive to both the spatial granularity (number of sub-districts) and poverty threshold used. Improvements in targeting accuracy, measured by reductions in inclusion and exclusion errors, are consistently observed in semi-urban districts with more than 15 sub-districts. In contrast, in rural and urban districts with fewer sub-districts, area-level features sometimes add noise rather than useful variation, resulting in negligible or even negative impacts on model performance. These findings suggest that a minimum level of spatial disaggregation is necessary for area-level features to meaningfully contribute to poverty targeting.

We also observed that classification errors are systematic rather than random. Poor households residing in wealthier or more urbanised areas are more likely to be excluded (misclassified as non-poor), particularly when features such as high NTL intensity, presence of private schools or built-up infrastructure signal affluence. Conversely, inclusion errors are more common in less-developed areas where non-poor households are misclassified as poor due to weak household-level indicators. Importantly, we found that the relationship between area-level features and poverty status is not always consistent across districts. For instance, while NDVI typically shows a negative association with household expenditure in semi-urban districts, in certain districts such as Sumenep, it correlates positively, likely due to the economic role of productive croplands. This highlights the context-specific nature of area-level features and the need for localised interpretation.

Overall, our findings underscore that area-level features can meaningfully enhance poverty targeting when combined with a context-aware modelling approach that accounts for spatial resolution and local socio-economic dynamics. Future work could explore a broader range of area-level indicators, such as road density, access to markets, mobile phone coverage, labour force participation, etc., which may offer complementary insights into household access, infrastructure and economic opportunity. However, we note that such features were not included in this study due to limited availability or inconsistent quality across the time periods examined. As more reliable and temporally consistent sources of these data become accessible, their integration into PMT models could further improve the precision, equity and contextual relevance of poverty targeting systems.

Another research direction is to use features from satellite imagery and administrative records with domain adaptation methods, TrAdaBoost.JQL [14] and WeightedMMoE [15], which we previously built to address distribution shifts in pooled survey data. We could treat districts that have many sub-districts and strong area-level features as “source” domains and let TrAdaBoost.JQL or WeightedMMoE learn which features and which source districts transfer best to low-granularity “target” districts where area-level features currently add noise. By streaming fresh satellite imagery and land cover data into a pipeline, the PMT model could update between surveys. This process could automatically correct for shifting light patterns

or new infrastructure. Further research could address extensions of the use of the Shapley value for machine learning applications with domain adaptation.

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Data availability The data that support the findings of this study are available from Statistics Indonesia, but restrictions apply to the availability of these datasets, which were used under licence for the current study and so are not publicly available. The datasets are, however, available from the authors upon reasonable request and with the permission of Statistics Indonesia.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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